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NSP/004/101 (OHI 1) Guidance on pole and stay holes.

1. Purpose

The purpose of this document is to describe the approved techniques for setting out the position of pole or stay holes and their subsequent excavation and foundation and backfilling requirements for use on the CE Electric UK Distribution System.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NSP/004/101	1 (May 06)	(OHI 1) Guidance on pole and stay holes
RTN/001/500/741/001	1.0	GN 74-01 Guidance note for pole and stay positions
RTN/001/500/741/002	1.0	GN 74-02 Guidance note for pole and stay hole excavations
RTN/001/500/741/003	1.0	GN 74-03 Guidance note for pole foundations
OHI 1	Oct 2000	Pole hole and stay excavations

2. Scope

This document has been prepared to provide guidance on the foundation and excavation requirements for all wood pole design specifications used within CE Electric UK.

3. Technical specification

3.1 Setting out pole holes

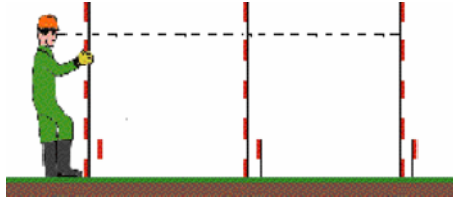
Checking the sighting of overhead line poles and associated stays prior to excavation is essential to make sure that poles are sited in the agreed line of travel and that stays are sited correctly to provide adequate stability of the pole to which they are attached.

All terminal and angle pole positions shall be marked accurately with two white wooden pegs by the surveyor. Straight line poles are also normally marked with a single white peg and poles must be accurately aligned between terminal/terminal, terminal/angle or angle/angle positions. Before starting any excavations, check the peg position, if any positional adjustments are required in excess of 150mm contact the job supervisor for advice before proceeding.

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3.1.1 Single poles

Using sighting rods, sight through from a terminal or an angle position to the next terminal or angle position. Check that all the pegs are in line with each other.



3.1.2 "H" poles

Using the same method as for single poles sight through and determine the line of the centre conductor. Measurements for the final pole positions will be taken from this line.

At an "H" pole position there may be one or two pegs. When there is only one peg this identifies the position of the centre conductor. When there are two pegs these identify the pole positions. **Note** that CE/C/37 section angle support poles are not sited equal distances from the centre conductor.

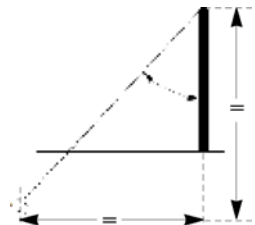
3.2 Setting out stay holes

Check the proposed stay positions. Make sure that terminal pole stays are in line with the conductors or splayed equally either side of this line and that angle pole stays are on the line of bisect or splayed equally either side of it. Where single, splayed, or tandem stay excavations are required then measurements can be made from one of these lines.

Remember that were possible the stay wire should be at a slope of 45° to the pole. To achieved this angle on level ground, plant the stay block the same distance away from the pole as the pole top make off is from the pole butt..

On sloping ground make allowances for the slope. In such cases use a 45° template and spirit level to check the stay angle.

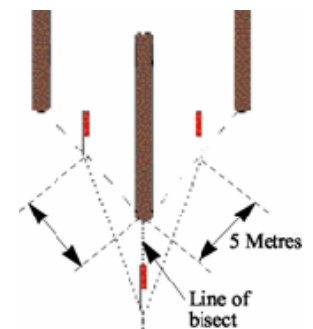
Additional guidance can be found in NSP/004/104.



Finding the line of bisect :-

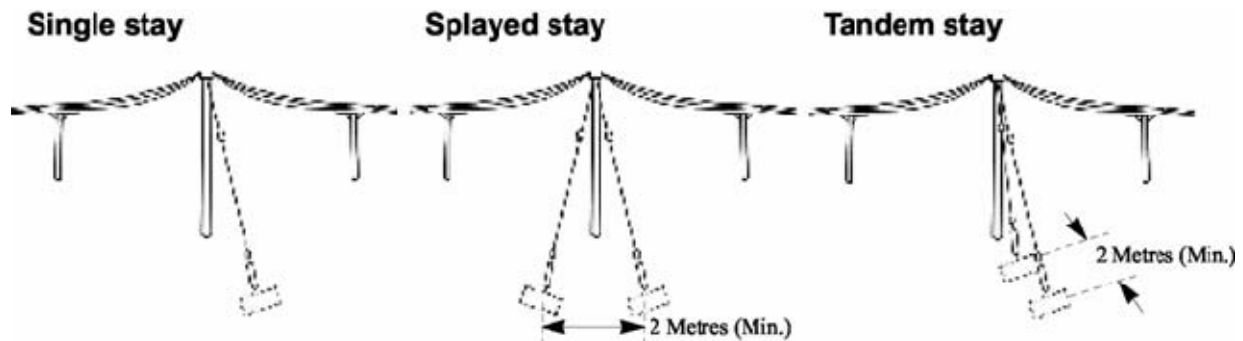
Knock a peg into the ground 5M from the angle pole centre and in line with the next pole centre. Repeat for the other side. Mark the centre of a 30M (approx) length of rope. At equal distances from each end of the rope tie the rope to the peg. Holding the rope at the centre mark, pull it tight forming a "V". Lay the rope on the ground and drive a peg into the ground adjacent to the rope centre mark.

A rope stretched between the pole centre and the peg will show the line of bisect. The diagrams below show the staying arrangements and stay excavation outlines for terminal and angle positions. Where more than one stay is used on a support



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it is important that each stay is equally tensioned. Further information on locating the correction position for different stay arrangements can be found on Drawing No 1091010273



3.3 Excavations

3.3.1 General

When excavating holes, the guidance given in WI 21-01 "Avoiding danger from underground Apparatus" shall be followed.

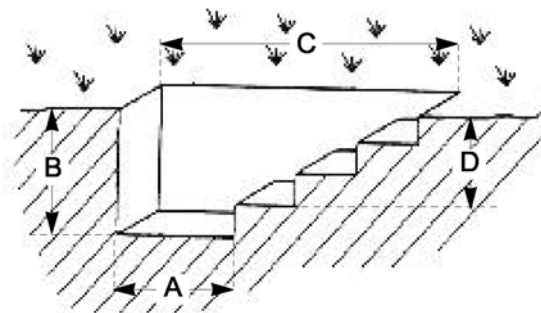
3.3.2 Single poles

Excavations for overhead lines poles and stays must be set out and excavated correctly to reduce the possibility of leaning or unbalanced poles.

Make sure that your excavations are to the correct depth and width as shown below. When using a mechanical excavator use a bucket with a width matched as near as possible to the size of the pole and follow the guidance given in GN 21-02 Using Mechanical Excavators.

It is important that wherever possible the excavation pit shall be in line with the line and as narrow as is practically possible. This will ensure that the pole will rest as near as possible against undisturbed ground and give maximum support against wind loading. The length of pole pit shall generally be equal to required depth. The excavation will be carried out in steps reaching full depth at the front of the hole. The ideal hole will therefore have a vertical front and side faces with a stepped rear. Any undercutting of the sides necessary to accommodate kicking blocks should be done with the removal of the minimum amount of soil from the faces of the hole.

- A = Diameter of butt + 150mm.
- B = Planting depth.
- C = Equal or greater than "B" dimension
- D = 1200mm



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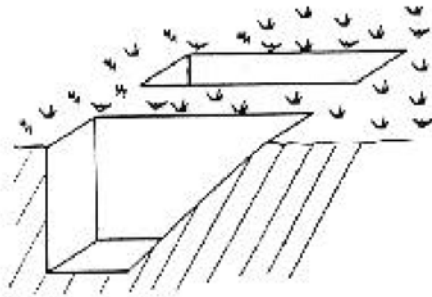
At straight line positions the peg will be left in and the pit excavated up to it so that on erection of pole it can be placed with its centre against the peg in order to reduce 'lining in' to a minimum.

At angle positions the pit must be carefully marked out around the peg so that the pole can be placed in the centre of the pit. Where ground conditions make accurate digging difficult, additional pegs should be put in line both directions for ease of 'lining in' after erection.

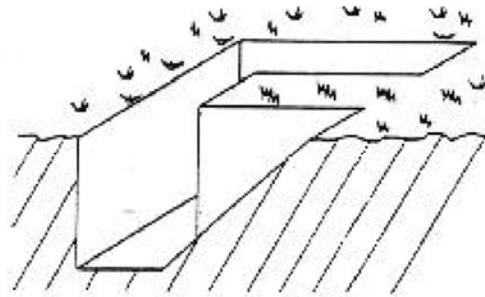
3.3.3 H, A or Rutter Poles

The side and length dimensions of the pole pit are governed by pole centres and foundation block dimensions plus tolerance each side to give adequate clearance during erection.

"H Pole mechanical excavation"

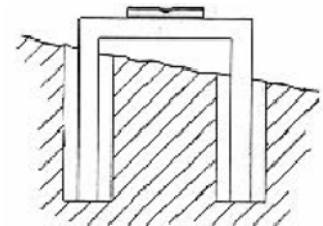


"H pole (with cross-member foundation baulks) mechanical excavation"



Where stability of ground will allow, pole erection will be easier if slotted trenches are cut and sloped into the pit in line with pole legs.

Where poles of 'Portal' type, with no cross foundation blocks, are to be erected, a single hole for each leg should be excavated, ensuring by means of a suitable template, that pit bottoms are at similar level.



3.4 Stay Hole Excavations

Stay blocks shall be undercut into firm ground and set for angle and terminal positions, as indicated on the line schedule. The stay pit shall be square with the line of the stay.

Excavation may be by hand or mechanical excavator. The sinking depth (vertical) for stay blocks shall be 1.6 m for lv lines. For hv lines, blocks shall be sunk to the depth specified on the appropriate drawing, No 1000439108 (Sheet 1 or sht 5) or 1091231166. The bottom of the stay pit must be undercut to accommodate the stay block so that pressure is taken against virgin ground and the stay rod must be slotted into the ground so that it falls in line with the stay. The stay rod must not be bent, the slot must be deep enough to accommodate it.



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The stay block must be held in a position in the undercut until sufficient soil has been placed and rammed in the bottom of the pit to hold it in position. The pit is then backfilled and reinstated in the manner described for pole pits.

Stay rods shall be in accordance with Drawing No 1000439101 sht 2.

Stay blocks shall be as quoted on the appropriate material list.

3.5 Backfilling

Strong foundations are essential to prevent poles falling over during normal working conditions or following conductor breakage. This is a very important operation and great care must be taken to ensure that excavated spoil is thoroughly compacted in order to hold the pole securely and reduce to a minimum future settlement.

All stay and pole foundations shall have the backfill material supplemented with "Perma Soil" soil stabiliser. A 25kg bag of Perma-Soil shall be used per pole leg. The powder shall be thoroughly mixed with the excavated pole hole spoil, half of this amount shall be used when the pole hole is being augured. Reinstatement/Compaction should be completed in lifts on approximately 300mm of material at a time. The mixed material must be placed in the pit and thoroughly rammed. After the initial 300mm layer the pole shall be checked for alignment and plumb with any correction being made at this time.

COSHH sheets are available for "Perma Soil" in the "Yellow Books"

Perma Soil is not poisonous, but it is an irritant and care should therefore be taken when handling it ie. Waterproof gloves should be worn when dispensing it.

25kg bucket, commodity code: 240777

25kg refill bag, commodity code: 305806

Where stoney ground is encountered, a layer of soil shall be placed in the pit first and after ramming, the stones strategically placed to key the pole to the sides of the pit.

To accommodate kicking blocks, the pit should be backfilled to the lower level of the baulk; alignment, twist and plumb checked and necessary excavation carried out to accommodate the kicking block before completion of backfilling as previously described.

Before leaving the pole or stay pit area, the surface must be reinstated, turf relaid etc and the whole area around the position cleared of all surplus spoil.

3.6 Pole Sinking Depths

3.6.1 LV Construction – NSP/004/041

With the exception of unstayed angle poles, all support foundations shall be sunk without kicking blocks at the depths shown:

Pole Height	Pole Sinking Depth
9.0 - 11.0m	1.6m
12.0 – 14.0m	1.8m

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3.6.2 HV Construction – NSP/004/042

Pole Height	Pole Sinking Depth Standard Block Founds Medium / Stout Poles	Augured Pole Foundations Medium Grade Poles	Augured Pole Foundations Stout Grade Poles
9.0 - 11.5m	1900 mm	2100mm	2400mm
12.0 - 14.0	2100 mm	2400mm	2600mm
15.0 - 18.0	2400 mm	2400mm	2600mm
19.0 - 22.0	3000 mm	Not Allowed	Not Allowed

3.6.3 EHV Construction (66 / 132kV) (OHL 4 and OHL 9)

Pole Height	Pole Sinking Depth
Upto 12.0m	2.0m
13.0 – 15.0m	2.2m
16.0 – 18.0m	2.4m
20.0 – 24.0m	3.0m

3.6.4 EHV Portal Construction (66 kV) (CE/C/37)

Pole Height	Pole Sinking Depth
<= 11.5m	1.8m
12.0 – 13.0m	2.0m
14.0 – 15.0m	2.1m
16.0 – 17.0m	2.25m
18.0 – 20.0m	2.4m

3.7 Pole Foundations

3.7.1 LV Construction – NSP/004/041

Unstayed angle supports shall be avoided wherever possible, however they are permitted when installed as follows:

ABC Main – Medium grade poles with a max angle of deviation of 7°

ABC Main – Stout grade poles with a max angle of deviation of 19°

Extended service spans – medium grade poles with a max angle of deviation 30°

In all cases the max pole height is limited to 12m and the poles must be sunk to a depth of 1.8m and fitted with 2 x ENA TS 43-91 Type 2 (1300 x 250 x 125)mm blocks located 500mm below ground level and attached with an M20 bolt. See drawing 1090431216 or 1091193332 for details.

Tubular steel supports shall be erected on a concrete slab 50 mm thick by 600 mm square. The planted section of the pole shall be surrounded with 100-300 mm of concrete, depending upon the load on the pole and ground condition, as Drawing No 1091193502.

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3.7.2 HV Construction – NSP/004/042 & NSP/004/044

Standard Pole Foundations

All poles shall be fitted with two wood foundation blocks (1300 x 250 x 125mm) as detailed in ENA TS 43-91 fig 3, Type 2 or company drawing 1000439103 sht1. The blocks shall be placed as follows:

Ground line to Top Block Pole Bolt = 500mm

Ground line to Bottom Block Pole bolt = 880mm.

Where ground conditions are exceptionally poor, the foundation, as shown on Drawing No 66.5/10.419, may be fitted.

Augured pole foundations

May be utilised as an alternative to standard wood block foundations detailed above. Care shall be taken during the installation of these poles to ensure that the auger sizes match as close as possible to the pole base diameter.

3.7.3 EHV Construction – (66kV) – OHL 4

Two Kicking blocks 1300 mm long to Drawing No 1000439103, Type 2 shall be fitted to single straight line and angle supports, except in the case of poles to Drawing No 1091231193, where a foundation baulk as specified on the drawing shall be fitted. Where conditions warrant it, additional foundation argumentation to Drawing No 1091231187, Sheet 2, is available for 'H' pole supports with 2.9 m centres. The latter will always be fitted to 'H' poles to Drawing No 1091231173 and 1174. Where necessary, all pole legs shall be erected on two 250 x 125 x 850 mm long creosoted wood baulks laid together to form a 500 x 850 mm footing.

3.7.4 EHV Construction – (66/ 132kV Single Circuit) – OHL 9

To be added

3.7.5 EHV Construction – (66kV Single Circuit) – CE/C/37

Two kicking blocks 1300 mm long to Drawing No 1000439103, Type 2 shall be fitted on portal straight line intermediate and section supports arranged to drawing arrangements Y104L0101 & Y104L0102. Section angle and terminal "H" structures arranged to drawing arrangements Y104L0103 – Y104L0109 shall utilise 2 x (2.44m centred) brace block fastened between the pole legs.

3.7.6 EHV Construction – (66kV Double Circuit) – CE/OM/DC1

Two kicking blocks 1300 mm long to Drawing No 1000439103, Type 2 shall be fitted to each leg of the portal rutter legs on the straight line intermediate and section supports arranged to Y105L0101 & Y105L0104. Section angle and terminal "H" structures shall utilise 2 x (2.44m centred) brace block fastened between the pole legs.

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4. References

4.1 External Documentation

Reference	Title
ENA TS 43-91	Stay Strands & Stay Fittings for Overhead lines

4.2 Internal documentation

Reference	Title
WI 21-01	Avoiding danger from underground Apparatus
GN 21-02	Using Mechanical Excavators
DSS/004/040	Specification for Yorkshire Electricity Overhead Lines to CE/C/37(M) and CE/OM/DC1(M)
NSP/004/041	Specification for Aerial Bundled Conductors (ABC) installed on new three phase low voltage overhead distribution systems
NSP/004/042	HV single circuit overhead lines on wood poles for voltages up to and including 33kV(In accordance with EATS 43-40 Ver 2.0)
NSP/004/044	Specification for Single Circuit Lines of Compact Covered Construction on Wood Poles for use at Voltages up to and including 33kV.
NSP/004/104.	Guidance on the types and installation requirements for stays
OHL 4	Specification For 66kV Overhead Lines on Wood Poles
OHL 9	Specification For 132kV Overhead Lines on Wood Poles

5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Line Standards Manager	G Hammel	04/01/07

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	08/01/07

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	08/01/07

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	03/01/07

6.5 Approval

Approval is given for the content of this document

		Sign	Date
D Sillito	Major Project Manager South	D Sillito	29/01/07
S Gardner	OHL Construction Manager	S Gardner	23/03/07

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
Neil Denley	Primary Engineering Projects Manager	Neil Denley	29/03/07